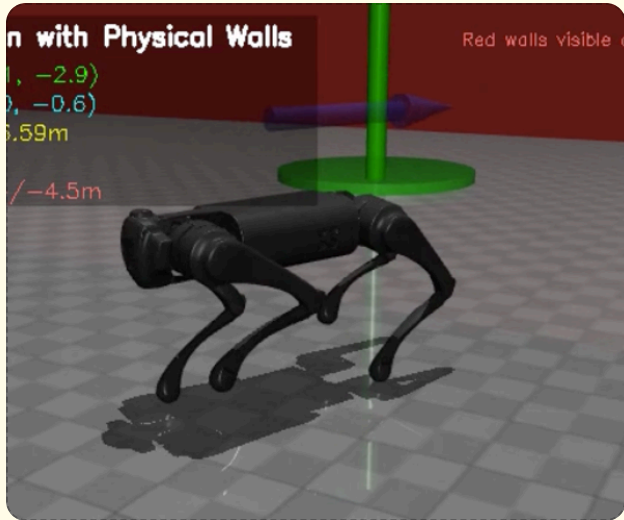


An AI Scientist Guided by Encoded Human Research Preferences

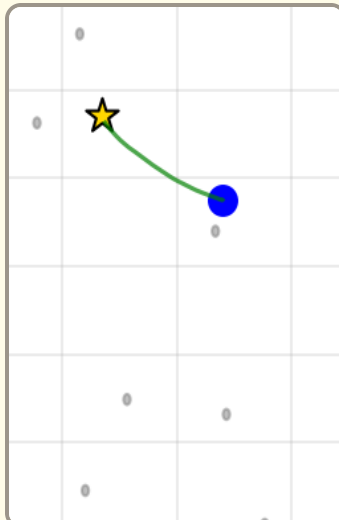
A Case Study in Quadrupedal Robot Neural Navigation Yiwen Zhang*, Eloise Zeng*, Jaeha Lee*, Tony Yue Yu (*equal contribution)

THE TASK — NAVIGATE TO THE GOAL THROUGH FIELDS UP TO 4× DENSER THAN DURING TRAINING

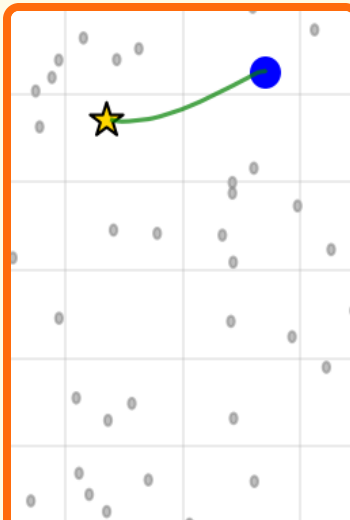
- Simulated quadruped (Unitree Go1, MuJoCo) must reach a goal marker by navigating through randomly generated obstacle fields
- Hierarchical learning:** a frozen locomotion policy walks; we train only the navigation layer on top, sensing with **64-beam LIDAR**
- Policies train at **one** obstacle density, then are scored from **0.25× to 4×** that density
- Composite score** (0–100) = success rate averaged over the six test densities, denser-than-training densities weighted 1.5–2.5×



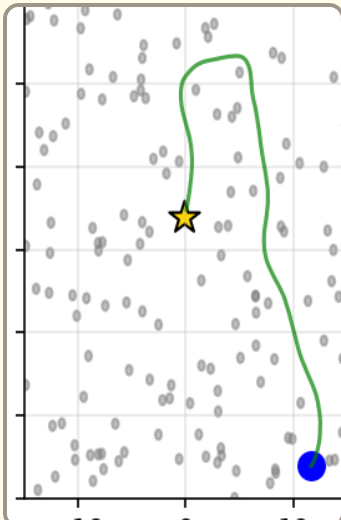
The robot



0.25×



Trained here



4× denser

THREE ATTEMPTS AT AN AI SCIENTIST

1. Unstructured agent

One free-running agent designed, ran, and graded its own experiments — it trained on the densities it was scored on and called failed predictions “partial wins”.

2. Structured agents

A fresh agent per step, and no agent grades its own work. Every experiment freezes its prediction and failure condition before running; verdicts are mechanical.

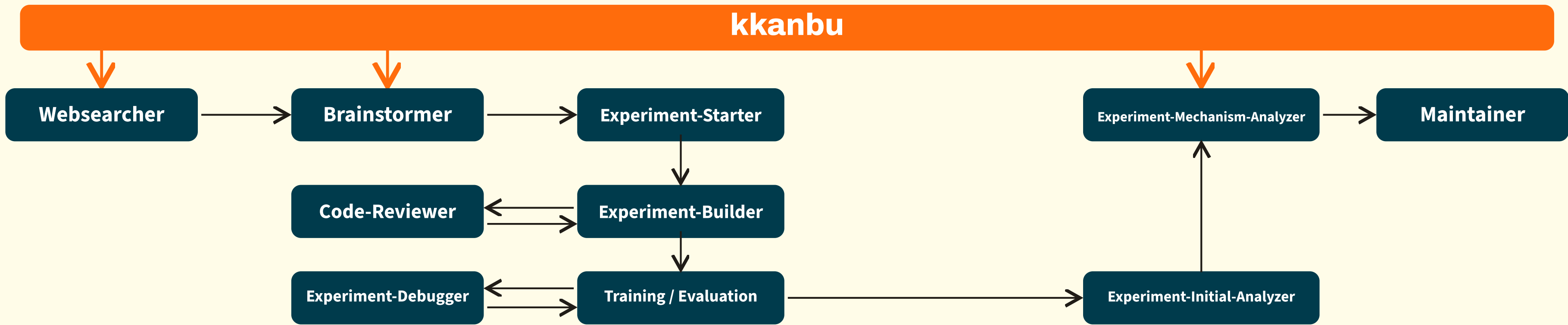
Best: 90.6

3. Structured agents + taste

Kkanbu: an MCP server exposing the researcher’s preferences as a knowledge graph — 57 claims, 86 relations, built through Socratic interviews — consulted at every direction-setting hand-off.

Best: 95.3 ★

EVERY EXPERIMENT IS ONE PASS THROUGH THIS LOOP — **TASTE IS CONSULTED AT FOUR HAND-OFFS; NO ONE ELSE HOLDS AN OPINION**



EXAMPLES OF KKANBU'S TASTE

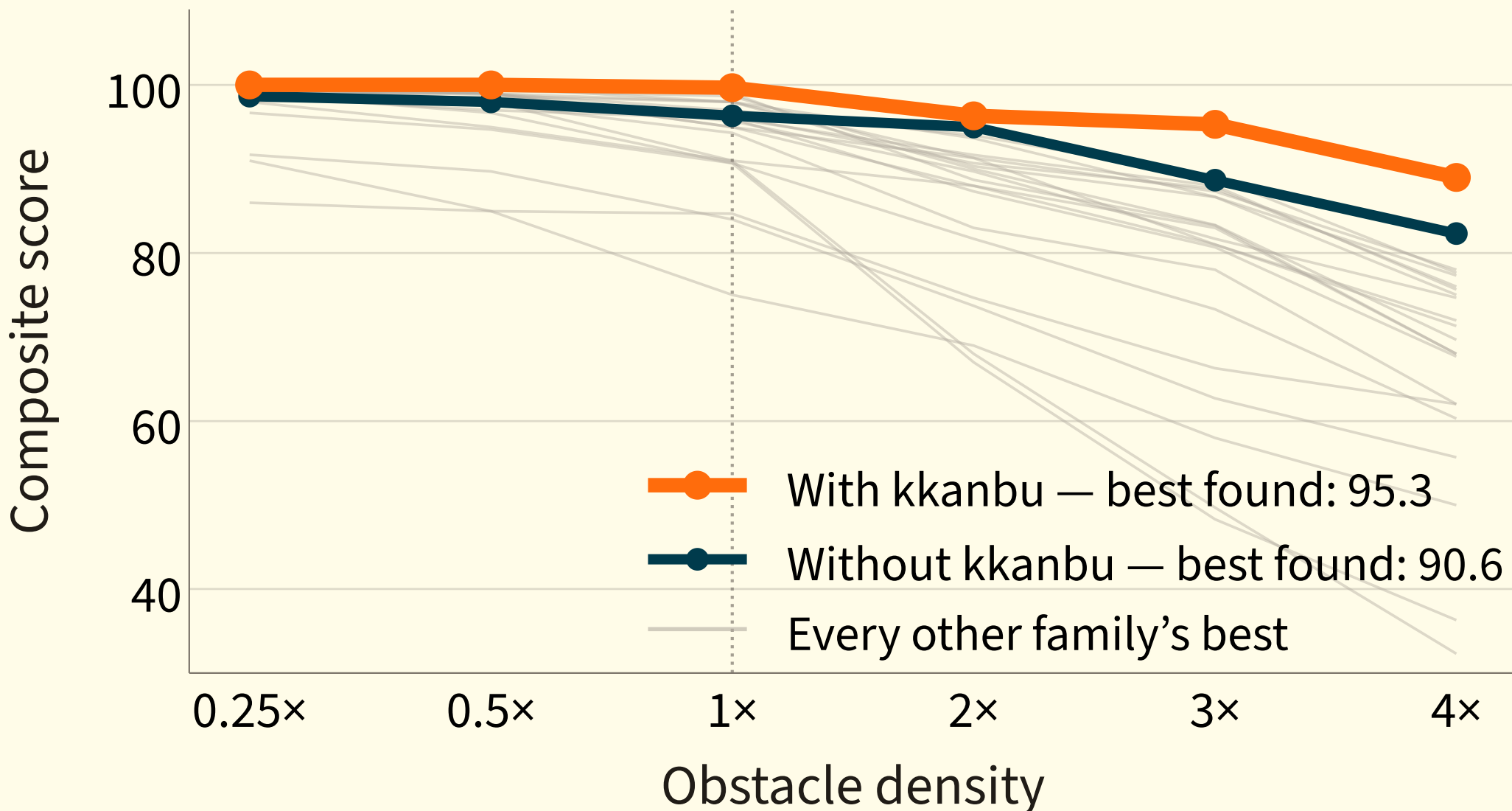
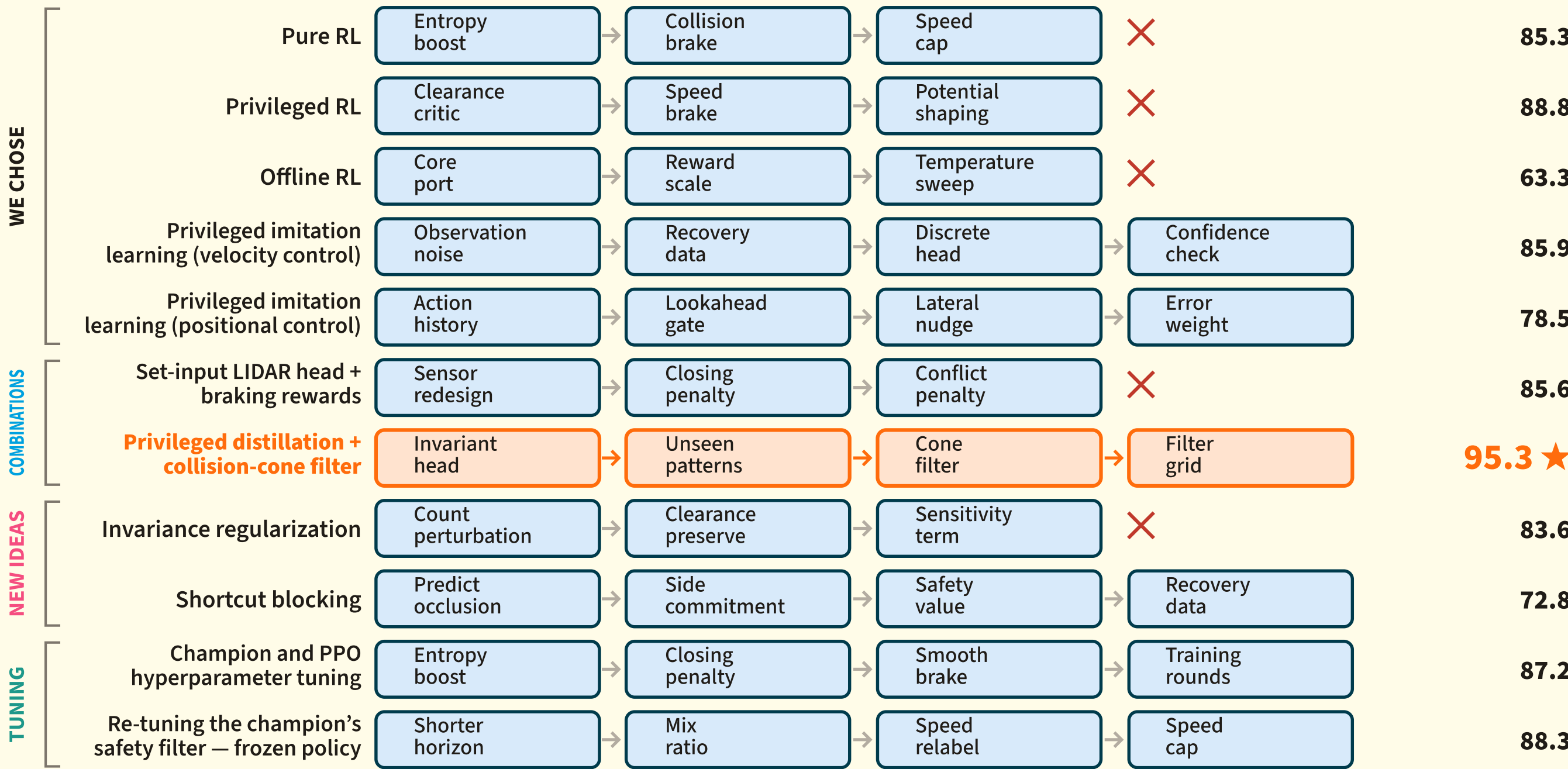
An experiment is only worth running if you learn something whether it succeeds or fails — otherwise it’s tuning, not research.

Breadth before depth: compare families of approaches on the same benchmark before committing to tuning one.

Freeze solved layers, and retire a paradigm once its bottleneck is proven to lie upstream.

11 MODEL FAMILIES — 5 WE CHOSE, 6 IT DESIGNED WITHIN OUR GUIDELINES. ONE CHAMPION.

Each box is one experiment; rows iterate left → right. ✕ marks a family retired early; ★ the champion.



SAME RULES. SAME TASK.

Kkanbu’s edge: sharper follow-up questions.

	WITH KKANBU	WITHOUT KKANBU
Experiments run	44 — kept asking	29 — quit early
Best model	95.3 ★	90.6
Champion recipe	Behavioral cloning of a privileged teacher + tuned collision-cone filter	Behavioral cloning on evasive demonstrations
Predictions proven wrong	35 of 44	21 of 29
Winning move	“Re-tune, don’t retrain” — evaluation-time filter tuning (its idea)	New demonstrations from a teacher that brakes & sidesteps
Stayed honest?	Yes	Yes

Structure keeps an AI scientist honest.
Taste decides where it looks.

